

5.

a) Total time to transfer

= Bus transfer time from
processor to device

+ data transfer time

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Device to processor

$$\Rightarrow 10000250 = 100 + \text{data transfer time} + 150$$

$$\begin{aligned}\Rightarrow \text{data transfer time} &= 10000000 \text{ ns} \\ &= 10000000 \times 10^{-9} \text{ s} \\ &= 0.01 \text{ s}\end{aligned}$$

So,

In 0.01s the device transfers 512 byte data

$$\therefore \text{ " 1 s " " " " } \frac{512}{0.01} \text{ " "}$$

$$= 51200 \text{ byte}$$

$$= \frac{51200}{1024} \text{ KB}$$

$$= 50 \text{ KB}$$

$$\therefore \text{ Transfer rate} = 50 \text{ KB/s}$$